

CHAPTER ONE

INTRODUCTION

1.1 Background to the Study

Sawmill operations are a common feature of semi-urban communities in Nigeria, where timber processing activities contribute to local economic development. These operations involve processes such as log transportation, wood cutting, sawing, planing, and sanding, which are carried out using heavy machinery including chainsaws, circular saws, and power generators (Biose *et al.*, 2025). While these activities are essential for wood production, they are also significant sources of environmental pollution, particularly noise and air pollution. Noise pollution in sawmill refers to the excessive and unwanted sound generated from machinery during wood processing, which can disrupt normal activities and pose health risks (Aletan and Garba, 2020). Prolonged exposure to high noise levels has been associated with hearing impairment, stress, and even cardiovascular disease (Munzel *et al.*, 2018).

Air pollution occurs when chemicals, particulate matter, or biological materials are released into the atmosphere in ways that can harm human health, cause discomfort to other living organisms, or damage both natural ecosystems and the built environment (Fagorite *et al.*, 2021). In sawmills, it arises from the release of wood dust and harmful gases such as carbon monoxide (CO), volatile organic compounds (VOCs), formaldehyde, and other hazardous substances into the atmosphere. These pollutants degrade air quality and have been linked to respiratory and cardiovascular diseases, as well as general health deterioration (Olalekan *et al.*, 2020).

The behaviour and impact of these pollutants are influenced by atmospheric conditions. Meteorological parameters such as temperature, wind speed, atmospheric pressure, and relative humidity play a crucial role in determining the dispersion, concentration, and movement of pollutants in the atmosphere. Variations in these conditions can affect pollutant accumulation and the level of exposure experienced by workers and nearby residents. Despite the environmental and health risks associated with sawmill activities, there is limited localized data on the combined assessment of noise pollution and air quality in semi-urban areas such as Ozoro, Delta State.

Ozoro is a semi-urban town located in the Isoko North Local Government Area of Delta State, Southern Nigeria. It serves as the administrative headquarters of the local government and is home to a growing population engaged primarily in agriculture, trade, and small-scale

commerce. The town hosts Southern Delta University, Ozoro, one of the state's key tertiary institutions, and has experienced steady urbanization driven by increased economic activity and population influx. Like many semi-urban communities in the Niger Delta region, Ozoro has seen a rise in informal industrial activities, including sawmilling, which have expanded alongside the growth of the built environment. Sawmill operations in Ozoro are carried out within or in close proximity to residential areas, a pattern common in Nigerian towns where land use planning has historically been inadequate (Aletan and Garba, 2020). These operations typically involve the use of chainsaws, circular saws, band saws, and power generators to process timber sourced from the surrounding forest belt of the Niger Delta, generating both noise and airborne pollutants as continuous by-products of daily production activity. Studies conducted in comparable sawmill environments within the Niger Delta region have consistently reported noise levels and gaseous pollutant concentrations in excess of permissible standards, with documented health consequences for both workers and nearby residents (Biose *et al.*, 2025). Despite this, there has been no recent assessment of noise pollution and air quality conducted specifically within the sawmill environment in Ozoro, this has left a significant gap in the environmental data available for this community. This study therefore aims to assess the levels of noise pollution and air quality in sawmill environment in Ozoro, with emphasis on selected pollutants and meteorological conditions.

1.2 Statement of the Problem

Sawmill operations are significant sources of environmental concern due to their continuous generation of noise and air pollutants during wood processing operations. The use of heavy machinery such as chainsaws, circular saws, and generators produces high levels of noise, while processes such as cutting, sawing, and sanding release wood dust and harmful gaseous emissions into the atmosphere. The persistent exposure to high noise levels in sawmill environments has been associated with adverse effects such as hearing impairment, stress, fatigue and loss of concentration. In addition, individuals within the vicinity of such operations may also be affected by prolonged noise exposure, which can disrupt daily activities and reduce quality of life (Biose *et al.*, 2025).

Furthermore, the emission of air pollutants such as carbon monoxide (CO), volatile organic compounds (VOCs), formaldehyde, chlorine gas, and hydrogen cyanide contributes to the deterioration of air quality. Continuous exposure to these pollutants has been linked to respiratory disorders, cardiovascular problems, eye irritation, and general health deterioration

among exposed individuals (Olalekan *et al.*, 2020). These environmental conditions pose serious risks to both workers and residents who may be exposed to such pollutants over time. Despite these environmental and health concerns, there is limited data on the combined assessment of noise pollution and air quality in sawmill environments of Ozoro, Delta State. This lack of information in the study area makes it difficult to evaluate the extent of pollution and develop appropriate environmental control measures. Therefore, there is a need for the assessment of noise pollution and air quality in the sawmill environment of Ozoro, in order to provide reliable data for effective environmental management and protection of public health.

1.3 Aim and Objectives of the Study

Aim of the Study

The aim of this study is to assess noise pollution and air quality at the sawmill environment in Ozoro, Delta State.

Objectives of the Study

The specific objectives of this study are to:

1. To determine the levels of noise pollution within the sawmill environment.
2. To determine the concentrations of selected air pollutants (carbon monoxide (CO), volatile organic compounds (VOCs), formaldehyde, chlorine gas, and hydrogen cyanide).
3. To compare the measured noise levels and air pollutant concentrations to established environmental permissible standards.
4. To proffer sustainable sawmilling practices within the study area

1.4 Research Questions

The following research questions will guide this study:

1. What are the levels of noise pollution within the sawmill environment?
2. What are the concentrations of selected air pollutants (carbon monoxide (CO), volatile organic compounds (VOCs), formaldehyde, chlorine gas, and hydrogen cyanide) within the sawmill environment?
3. To what extent do the observed noise levels and air pollutant concentrations comply with established environmental permissible standards?
4. What are the sustainable sawmilling practices within the study area?

1.5 Justification of the Study

Ozoro, as a growing semi-urban community, hosts active sawmill operations that function in close proximity to residential zones and an educational institution. Despite this, there has been no recent assessment of noise pollution and air quality conducted specifically within the sawmill environment in Ozoro. This gap is significant given that sawmill workers and nearby residents in Ozoro are potentially exposed to a combination of hazardous conditions on a daily basis, without any recent empirical evidence to quantify the extent of that exposure.

Studies conducted in comparable sawmill environments across Nigeria have documented exceedances of regulatory standards for both noise and gaseous pollutants, with particular concern over carbon monoxide, VOCs, and particulate matter (Olalekan *et al.*, 2020; Biose *et al.*, 2025). However, the specific pollutant profile of the Ozoro sawmill environment, including gaseous emissions such as formaldehyde, chlorine gas, and hydrogen cyanide, has not been characterized. Without this baseline information, it is not possible to determine whether the current conditions in Ozoro comply with established national and international environmental standards or what level of risk is being imposed on exposed individuals.

Furthermore, the influence of meteorological conditions on pollutant behaviour in this specific environment is unexplored. Factors such as wind speed, temperature, and humidity affect how noise propagates and how airborne pollutants disperse, and their role cannot be inferred from studies conducted in different geographical settings. This study therefore addresses a clear and specific knowledge gap by providing original, location-specific data on noise levels, air pollutant concentrations, and meteorological parameters at a sawmill in Ozoro. The findings are intended to serve as a primary reference for environmental regulators, public health practitioners, and policymakers working to mitigate industrial pollution in semi-urban communities in Delta State and the wider Niger Delta region.

1.6 Significance of the Study

This study is significant as it provides empirical data on the levels of noise pollution and air quality in a sawmill environment in Ozoro, Delta State. The findings will contribute to a better understanding of the extent of environmental pollution associated with sawmill operations, particularly in semi-urban areas where such operations are often located close to residential areas. The results of this study will be beneficial to environmental regulatory agencies by providing baseline information that can support monitoring, policy formulation, and enforcement of environmental standards. It will also assist public health authorities in

identifying potential health risks associated with prolonged exposure to noise and air pollutants such as carbon monoxide, volatile organic compounds, and other hazardous gases commonly emitted in sawmill environments.

Furthermore, the study will be valuable to sawmill operators and the workers by creating awareness of the potential environmental and health implications of their activities, thereby encouraging the adoption of safer and more sustainable practices. It will also serve as a reference material for researchers and students in the field of Environmental Management and Toxicology, contributing to existing literature on industrial pollution and its impacts.

In addition, the inclusion of meteorological parameters in this study will enhance understanding of how environmental conditions influence pollutant dispersion, thereby providing a more comprehensive assessment of environmental quality in the study area. Overall, the findings of this study are expected to support efforts aimed at improving environmental quality and safeguarding public health in Ozoro and similar communities.

1.7 Scope of the Study

This study focuses on the assessment of noise pollution and air quality within a selected sawmill environment in Ozoro, Delta State. The study specifically involves the measurement of noise levels and selected air pollutants (carbon monoxide (CO), volatile organic compounds (VOCs), formaldehyde, chlorine gas, and hydrogen cyanide). In addition, the study considers key atmospheric meteorological parameters such as temperature, wind speed, atmospheric pressure, and relative humidity in order to evaluate their influence on the distribution and concentration of pollutants within the study area. Data collection will be carried out through direct field measurements at designated sampling points within the sawmill, as well as at a control location with minimal anthropogenic activities for comparative purposes.

The study is limited to measurements conducted at specific periods of the day and repeated three times to ensure consistency and reliability of data. It does not cover other potential environmental pollution such as soil or water contamination, nor does it include detailed medical or clinical assessment of health conditions among workers or residents. Furthermore, the study is geographically restricted to the selected sawmill and its immediate surroundings in Ozoro, and as such, the findings may not be generalized to all sawmill environments without consideration of local variations.

1.8 Limitations of the Study

1.9 Definition of Terms

1. Volatile Organic Compounds (VOCs): A group of organic chemicals that easily evaporate into the air and may have harmful health effects when present in high concentrations.
2. Formaldehyde: A colorless, strong-smelling gas commonly released from wood processing activities and known to have adverse health effects.
3. Carbon Monoxide (CO): A colorless, odorless gas produced from incomplete combustion processes, which can be harmful when inhaled in high concentrations.
4. Chlorine Gas: A toxic gas that may be present in industrial environments and can cause respiratory irritation upon exposure.
5. Hydrogen Cyanide: A highly toxic gas that can interfere with the body's ability to use oxygen effectively.
6. Control Location: A site selected for comparison, characterized by minimal human or industrial activities, used to determine background environmental conditions.
7. Sawmill Environment: The area within and around the sawmill where wood processing activities take place and where environmental measurements are conducted.